
Concept Questions: Chapter 4

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ME-3640 – Heat Transfer

2026-Spring



Abstract

Fundamental principals and concepts are far more important to engineering expertise than understanding of example systems. These concept questions (shamelessly lifted from the textbook) will help you test your knowledge after reading the chapter summary and building a concept map. Use the relative ease and difficulty you experience while answering these questions to prioritize your efforts for the remainder of this chapter.

1 Lumped System Analysis

- 4-1C** What is the physical significance of the Biot number? Is the Biot number more likely to be larger for highly conducting solids or poorly conducting ones?
- 4-2C** What is lumped system analysis? When is it applicable?
- 4-3C** In what medium is the lumped system analysis more likely to be applicable: in water or in air? Why?
- 4-4C** For which solid is the lumped system analysis more likely to be applicable: an actual apple or a golden apple of the same size? Why?
- 4-5C** For which kind of bodies made of the same material is the lumped system analysis more likely to be applicable: slender ones or well-rounded ones of the same volume? Why?
- 4-6C** Consider heat transfer between two identical hot solid bodies and the air surrounding them. The first solid is being cooled by a fan while the second one is allowed to cool naturally. For which solid is the lumped system analysis more likely to be applicable? Why?
- 4-7C** Consider heat transfer between two identical hot solid bodies and their environments. The first solid is dropped in a large container filled with water, while the second one is allowed to cool naturally in the air. For which solid is the lumped system analysis more likely to be applicable? Why?
- 4-8C** Consider a hot baked potato on a plate. The temperature of the potato is observed to drop by 4°C during the first minute. Will the temperature drop during the second minute be less than, equal to, or more than 4°C ? Why?
- 4-9C** Consider a potato being baked in an oven that is maintained at a constant temperature. The temperature of the potato is observed to rise by 5°C during the first minute. Will the temperature rise during the second minute be less than, equal to, or more than 5°C ? Why?
- 4-10C** Consider two identical 4kg pieces of roast beef. The first piece is baked as a whole, while the second is baked after being cut into two equal pieces in the same oven. Will there be any difference between the cooking times of the whole and cut roasts? Why?
- 4-11C** Consider a sphere and a cylinder of equal volume made of copper. Both the sphere and the cylinder are initially at the same temperature and are exposed to convection in the same environment. Which do you think will cool faster, the cylinder or the sphere? Why?

2 Transient Heat Conduction in Large Plane Walls, Long Cylinders, and Spheres with Spatial Effects

4-39C An egg is to be cooked to a certain level of done-ness by being dropped into boiling water. Can the cooking time be shortened by turning up the heat and bringing water to a more rapid boiling?

4-40C What is an infinitely long cylinder? When is it proper to treat an actual cylinder as being infinitely long, and when is it not? For example, is it proper to use this model when finding the temperatures near the bottom or top surfaces of a cylinder? Explain.

4-41C What is the physical significance of the Fourier number? Will the Fourier number for a specified heat transfer problem double when the time is doubled?

4-42C Why are the transient temperature charts prepared using non-dimensionalized quantities such as the Biot and Fourier numbers instead of the actual variables such as thermal conductivity and time?

4-44C How can we use the transient temperature charts when the surface temperature of the geometry is specified instead of the temperature of the surrounding medium and the convection heat transfer coefficient?

4-45C The Biot number during a heat transfer process between a sphere and its surroundings is determined to be 0.02. Would you use lumped system analysis or the transient temperature charts when determining the midpoint temperature of the sphere? Why?

4-46C A body at an initial temperature of T_i is brought into a medium at a constant temperature of T_∞ . How can you determine the maximum possible amount of heat transfer between the body and the surrounding medium?

3 Transient Heat Conduction in Semi-Infinite Solids

4-82C Under what conditions can a plane wall be treated as a semi-infinite medium?

4-83C What is a semi-infinite medium? Give examples of solid bodies that can be treated as semi-infinite mediums for heat transfer purposes.

4-84C Consider a hot semi-infinite solid at an initial temperature of T_i that is exposed to convection to a cooler medium at a constant temperature of T_∞ , with a heat transfer coefficient of h . Explain how you can determine the total amount of heat transfer from the solid up to a specified time t .

A Rubrics

Submissions of this assignment will be graded upon the criteria found in this section. Review these carefully, and complete your work so that application of this rubric produces your desired score.

 Criterion 1 Problems	
<i>Homework is completed with appropriate effort, neatness, etc.</i>	
Points	Description
20	Honest effort to participate and learn: every problem shows an answer or questions which need resolution.
10	Rushed or haphazard work: some problems show no effort at all.
0	Work is absent, wholly incorrect, or plagiarized.